Analytical Biochemistry 690 (2024) 115491

Contents lists available at ScienceDirect

Analytical Biochemistry

journal homepage: www.elsevier.com/locate/yabio

MMDB: Multimodal dual-branch model for multi-functional bioactive peptide prediction

Yan Kang [a,b], Huadong Zhang [a], Xinchao Wang [a], Yun Yang [a,b,*], Qi Jia [c]

[a] National Pilot School of Software, Yunnan University, Kunming, 650091, Yunnan, China
[b] Yunnan Key Laboratory of Software Engineering, China
[c] School of Information Science, Yunnan University, Kunming, 650091, Yunnan, China

ARTICLE INFO

Keywords:
Multi-functional bioactive peptide prediction
Multimodal learning
Multi-scale dilated convolution
Peptide sequence
Peptide structural information

ABSTRACT

Bioactive peptides can hinder oxidative processes and microbial spoilage in foodstuffs and play important roles in treating diverse diseases and disorders. While most of the methods focus on single-functional bioactive peptides and have obtained promising prediction performance, it is still a significant challenge to accurately detect complex and diverse functions simultaneously with the quick increase of multi-functional bioactive peptides. In contrast to previous research on multi-functional bioactive peptide prediction based solely on sequence, we propose a novel multimodal dual-branch (MMDB) lightweight deep learning model that designs two different branches to effectively capture the complementary information of peptide sequence and structural properties. Specifically, a multi-scale dilated convolution with Bi-LSTM branch is presented to effectively model the different scales sequence properties of peptides while a multi-layer convolution branch is proposed to capture structural information. To the best of our knowledge, this is the first effective extraction of peptide sequence features using multi-scale dilated convolution without parameter increase. Multimodal features from both branches are integrated via a fully connected layer for multi-label classification. Compared to state-of-the-art methods, our MMDB model exhibits competitive results across metrics, with a 9.1% Coverage increase and 5.3% and 3.5% improvements in Precision and Accuracy, respectively.

1. Introduction

Bioactive peptides play a pivotal role in modulating various physiological functions [1], including but not limited to antimicrobial [2], anticancer [3], anti-inflammatory [4], and blood pressure regulation [5]. When ingested by individuals, bioactive peptides have demonstrated the ability to induce beneficial effects on human well-being, serving as a nutritional source and exhibiting a wide range of potential physiological functions within the organism [6]. These functions encompass a wide range of activities, including antimicrobial, antithrombotic, antihypertensive, and immunomodulatory effects [7]. Recently, there has been a growing identification of a diverse range of bioactive peptides exhibiting multifunctionality. As an example, Temporin is a short peptide secreted by frogs, displaying antimicrobial properties. Furthermore, recent research has indicated its potential as an agent against cancer and viruses [8]. In order to deepen the understanding of the mechanisms underlying the actions of bioactive peptides and meet the increasing demand for the development of natural foods and drugs, there has been a heightened interest in predicting the functionality of peptides. This enthusiasm has propelled advancements in diverse computational approaches.

In the literature, various computer-based methods utilizing physicochemical features have previously been proposed for peptide prediction. These methods predominantly leverage physical and chemical characteristics in peptide prediction. Essential features encompass pseudo amino acid composition (Pse-AAC), normalized amino acid composition (NAAC), hydrophobicity, net charge, isoelectric point, α-helical tendency, β-folding tendency, and steering tendency, among others. Machine learning and deep learning-based methods have been presented in the past decade. Among machine learning methods, PPCT [9] as a new feature descriptor has been developed to accurately identify nucleic acid-binding proteins by combining PSSM profiles and PSFM profiles. The StackCPPred [10] classifier employs the Pse-AAC feature, which possesses excellent structural properties, to predict cell-penetrating peptides. The TPpred-ATMV [11] method extracts 14 physicochemical features for peptide characterization. The PPTPP [12]

* Corresponding author. National Pilot School of Software, Yunnan University, Kunming, 650091, Yunnan, China.
E-mail address: yangyun@ynu.edu.cn (Y. Yang).

https://doi.org/10.1016/j.ab.2024.115491
Received 12 November 2023; Received in revised form 21 January 2024; Accepted 19 February 2024
Available online 7 March 2024
0003-2697/© 2024 Published by Elsevier Inc.

method introduces a new feature representation scheme and training process to produce and rank features associated with chemical and physical properties. Among deep learning methods, nuclear magnetic resonance is combined with structural descriptors to predict membrane-inhibiting peptides [13]. More recently, the StaBle-ABPpred model has been utilized to integrate peptide-level features (PELF) and amino acid-level features (AALF) for the prediction of antimicrobial peptides [14].

Recently, several prediction models have been developed to predict peptide function solely based on sequence information without utilizing experimental data or pre-existing information. Among machine learning-based models, the ACPred-Fuse [15] model integrates totaling 29 different sequence-based characteristic descriptors to explore the prediction of anticancer peptides. An IFFC-based model is designed to identify signal peptides based on peptide sequence information [16]. A comprehensive predictive model is based on the ensemble method and random forest to forecast inflammatory peptides induced by proinflammatory stimuli [17]. A pioneering predictor, AntiCP, integrates classical amino acid and dipeptide representation features with an SVM-based classifier for the detection of anticancer peptides [18]. Several popular classifiers have been presented among deep learning-based models by utilizing amino acid sequences of peptides. For instance, iACP-MultiCNN [19] develops the multi-channel CNN model utilizing protein sequences to identify anticancer peptides. AMPFinder is a cascaded computational model to identify AMPs and their functional type based on sequence-derived and life language embedding [20]. Deep-ABPpred is utilized for the identification of antimicrobial peptides in protein sequences by utilizing a bidirectional LSTM with word2vec [21]. Multimodal data fusion plays a pivotal role in biomedical data analysis, surmounting the constraints inherent to individual modalities, notably evident in the widespread application within neuroimaging through sophisticated multimodal fusion mechanisms [22–24]. Nevertheless, challenges emerge in the realm of multimodal data fusion, encompassing tasks such as preprocessing varied data sources, addressing data heterogeneity, and managing computational complexity [25–27]. Recently, deep learning methods such as enhanced CNN and generative adversarial networks have been presented to effectively tackle these challenges [28–30].

With the discovery of bioactive peptides with multiple functionalities, research on the functional prediction of multi-functional bioactive peptides has become increasingly valuable. However, predicting the complex and diverse functionalities of bioactive peptides remains a challenging task as there are limited computational methods available for predicting their functions, and there is an issue of low accuracy. A convolutional deep learning model, MLBP, is based on bioactive peptide sequences and a GRU-based deep learning method for the multi-functionality prediction of bioactive peptides [31]. However, due to MLBP only utilizing the sequence features of peptides without considering their structural features, its prediction accuracy needs to be further improved.

In summary, physicochemical characteristics have been shown to model the functional information of peptides effectively. However, previous studies on the prediction of multi-functional bioactive peptides only consider sequence vectors without utilizing physicochemical characteristics, thus failing to capture the complementary between diverse modalities. There is also a need to design more effective feature extraction methods to capture diverse and complex features. To address the above issues, we design a multimodal dual-branch (MMDB) model that combines peptide sequence vectors with structural keys (MACCS Key) from the physicochemical space for the multi-functional prediction of bioactive peptides from different modalities and perspectives. Specifically, the multi-scale dilated convolution with the Bi-LSTM branch adopts a representation learning model to extract complex sequence features from the sequence vector while the multi-layer convolution branch captures the structural feature from the structural key. Since dilated convolutions effectively capture diverse contextual information in the field of visual perception [32], we design multi-scale dilated convolution to the field of multi-functional bioactive peptide prediction without increasing parameters. The main contributions of this study are as follows:

- [•] Unlike most studies on multi-functional bioactive peptide prediction that depend on sequence information solely, we propose a novel lightweight deep learning model, MMDB, to effectively capture the complementary information of peptide sequences and structural properties of bioactive peptides.
- [•] The MMDB model employs two branches to capture multimodal characteristics from different perspectives. To the best of our knowledge, it is the first time we have applied dilated convolution for effectively mining features of peptide sequences at multiple scales by enlarging the receptive field.
- [•] Compared to state-of-the-art methods, our MMDB model demonstrates significant improvements in Precision, Coverage, and Accuracy by 3.7%, 9.1%, and 3.5%, respectively. And lots of ablation studies have been implemented to evaluate the performance of diverse modules of the model.

2. Methods

2.1. Datasets

To ensure the effective training and validation of our model, we utilize the same dataset as MLBP which is compiled by searching for bioactive peptides, including encompassing anti-microbial peptides (AMPs) [33], anti-cancer peptides (ACPs) [34], anti-diabetic peptides (ADPs) [35], antihypertensive peptides (AHPs) [36], and anti-inflammatory peptides (AIPs) [37], respectively. AIPs, AHPs, ADPs, ACPs, and AMPs are then extracted from PreAIP [38], mAHTPred [39], BioDADPep [40], AntiCP 2.0 [41], and AMPfun [42], respectively. Furthermore, the CD-HIT program is employed to obtain a total of 646 ACPs, 514 ADPs, 868 AHPs, 1678 AIPs, and 2409 AMPs by eliminating sequences with a similarity greater than 90%, thereby avoiding sequence redundancy and homology bias.

Bioactive peptides, which are natural compounds comprising amino acids connected by covalent linkages, exhibit a wide range of biological activities. Bioactive peptides exhibit diverse modes of action, AMPs, ADPs, AHPs, AIPs, ACPs, antioxidant, and immunomodulatory activities, among others [7,42]. Because of their antibacterial, antifungal, or antiviral properties, AMPs, also known as host defense peptides (HDPs), are widely distributed in the innate immunity system. When a host is invaded by a pathogen such as a virus or bacteria, AMPs are induced to eliminate or eradicate the invading pathogen through a membrane-disrupting mechanism [43,44]. AHPs exert their anti-hypertensive effects mainly through the inhibition of the angiotensin-converting enzyme (ACE), a critical regulator in the renin-angiotensin system (RAS) responsible for maintaining blood pressure and electrolyte balance. The modulation of this system by AHPs plays a role in maintaining cardiovascular homeostasis [45]. AIPs are naturally produced and have the ability to suppress T(H)1-driven reactions specific to particular antigens [46]. ACP inhibits tumor cell proliferation or migration, blocks tumor vascularization, and is unlikely to cause drug resistance [47]. Antioxidant peptides possess a free radical-neutralizing capability and exert their influence on cellular entities by eliminating reactive oxygen species, restraining oxidative damage to lipids, and modulating metal ion interactions [48]. An 80% portion of the peptides was randomly chosen as the training dataset, with the remaining 20% assigned as the validation dataset.

2.2. Problem formulation

Given a set of multi-modalities of peptide dataset $\mathscr{P} = \{p_0, p_1, \ldots, p_N\}$ of N instances, the problem aims to predict multi-function $\mathscr{F} : (p_k,$

$\theta) \rightarrow \widehat{y}_k$, where θ represents the parameters of the model, $p_k \in \mathscr{P}$, and $\widehat{y}_k$ is the functions of p_k. For each peptide p_k, the goal is to obtain a vector $\widehat{y}_k = \{\widehat{y}_{k1}, \ \widehat{y}_{k2}, \ldots, \widehat{y}_{kn}\}$ which $\widehat{y}_{ki}$ indicates whether the k_{th} peptide has the i_{th} function. Here, $p_k = \{(p_i^{\Delta}, p_i^{\Theta}) | p_i^{\Delta} \in \mathscr{P}^{\Delta}, p_i^{\Theta} \in \mathscr{P}^{\Theta}\}$ is peptide dataset with two modalities, where p_i^{Δ} is the sequence data and p_i^{Θ} is the structural data. Actually, the multi-functional prediction problem is a multi-label classification problem.

2.3. Multimodal data input

This study introduces two distinct representations of information, namely sequence vector and structural key. Two separate modalities of data input have different data structure characteristics as representational ideographs and information.

2.3.1. Peptide sequence data input

Sequence vectors are represented in the FASTA format, a text-based format used to depict nucleic acid or polypeptide sequences. In this format, amino acids are encoded by leveraging a single-letter notation for polypeptides. The embedding layer served as a crucial component in converting textual sequences into continuous numerical vectors. Before applying the embedding technique, we processed data on the amino acid sequences to ensure their suitability for further analysis. Since multi-functional bioactive peptides have different sequence lengths ranging from 5 to 517, we fill the peptides with specific features '0' with less than 517 residues. Thus, all features of the peptide are converted to integers. Ultimately, the sequence is encoded into a 517-dimensional representation to capture its essential characteristics.

2.3.2. Peptide structural data input

In this study, the structural key of multi-functional bioactive peptides is extracted by utilizing molecular fingerprints, which are represented as a binary vector of 0 and 1. In the present investigation, we utilize the RDkit software to transform the protein fragment's FASTA representation into an MACCS fingerprint based on the chemical structure fragments. MACCS key is composed of 166 bits of code, and each bit represents a predefined substructure (chemical structure fragment). If the substructure exists in a molecule, the corresponding bit of the molecular fingerprint is 1; otherwise, it is 0.

2.4. Model construction

The overall architecture of the MMDB model is given in Fig. 1, which consists of a multimodal input module, encoder module, and classification module. The input of multiple modalities is subsequently converted into a continuous representation by the embedding module. The encoder module is composed of multi-scale dilated convolution with Bi-LSTM branch and multi-layer convolution branch to extract features from sequence vector and structural key. The classification model final outputs the probability of the corresponding type of peptide.

2.4.1. Multi-scale dilated convolution with Bi-LSTM branch

We devise a novel multi-scale dilated convolutional and Bi-LSTM branch to enable the effective extraction of diverse characteristics within the peptide sequence while keeping the speed of lightweight networks fast and the amount of computation small. After the multi-scale dilated convolution, the data matrix contains the local information embedding of the sequence at different scales. Then, the data matrix

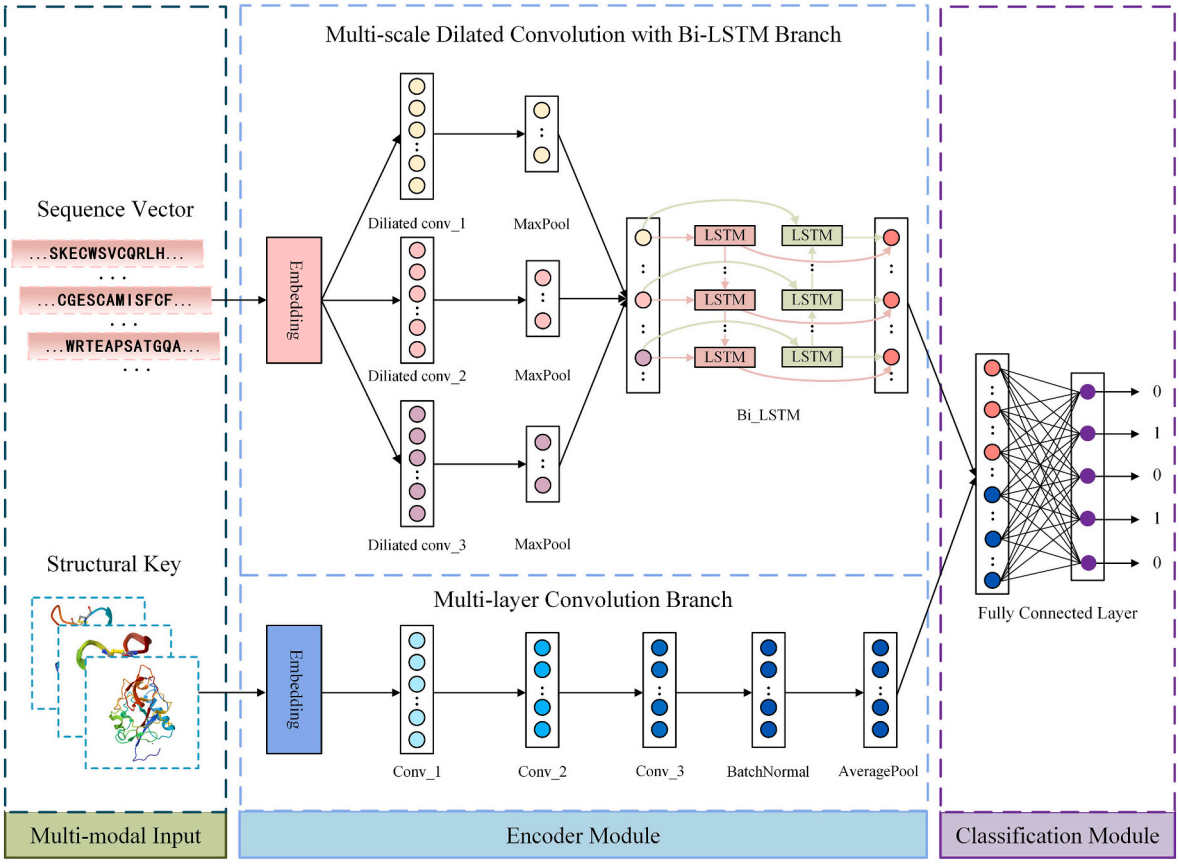


Fig. 1. Overview of MMDB model. The MMDB algorithm adopts a dual-branch architecture for multi-label prediction of multi-functional bioactive peptides. The multi-scale dilated convolution with the Bi-LSTM branch employs a multi-scale dilated convolution technique to extract peptide sequences' multi-scale information. In the multi-layer convolution branch, CNN is utilized for embedding and feature extraction of structural keys. The output of both branches is integrated into the classification module.

is fed into Bi-LSTM, which fully captures the distance dependence between local features by taking full advantage of the characteristics between multi-scale data and fusing multi-scale local information embedding into global embedding to extract more compelling features.

2.4.1.1. Multi-scale dilated convolution. Dilated convolution extends the receptive field without requiring an increase in network parameters [49]. Compared with traditional convolution, dilated convolutions provide a wider receptive field while maintaining resolution and coverage, offering higher computational and parameter efficiency. Presently, dilated convolutions find extensive application in domains including computer vision [50] and speech processing [51].

In this study, we design a multi-scale dilated convolution model, where three dilated convolutional layers are connected to the embedding layer parallelly as shown in Fig. 2. In order to extract diverse scales of contextual information from peptide sequences, we modulate different dilation rates of dilated convolutional layers separately. To prevent the loss of detailed information due to excessively large dilation rates and to avoid the integration of an excessive number of dilation rates, which might result in redundant information within the multi-scale dilated convolution model, we utilize three different dilation rates: 2, 4, and 8. At the same time, the structure of this multi-scale dilated convolution model is capable of extracting features from various receptive fields without increasing network parameters. When utilized in a one-dimensional convolutional neural network (1D CNN), this can be computed as follows:

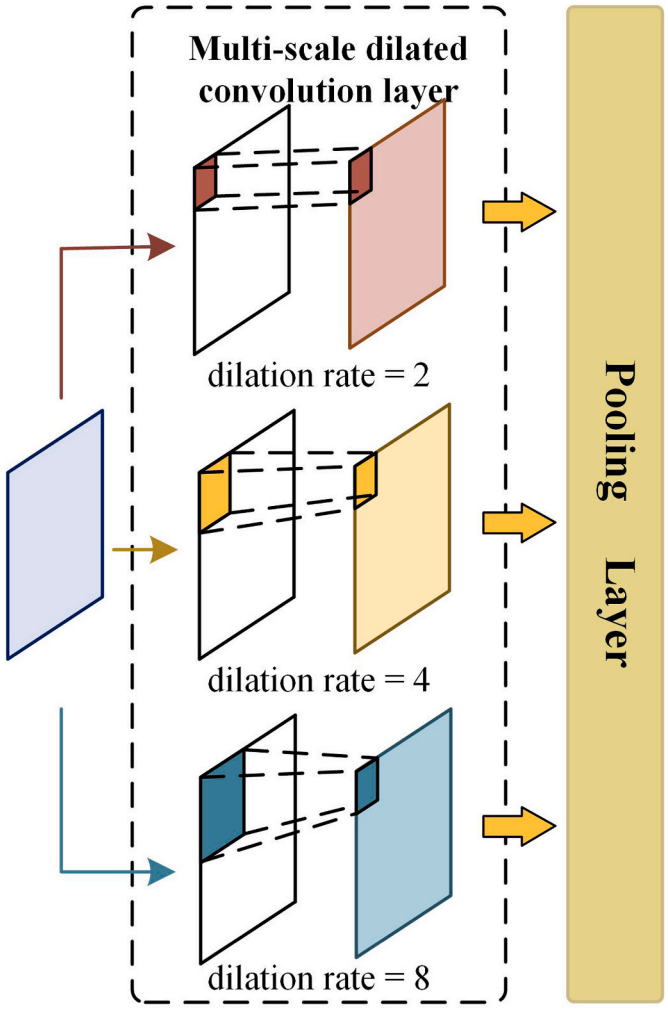


Fig. 2. Overview of multi-scale dilated convolution model.

$$y_i = \varphi\left(\sum_{k=1}^{K} p_{i+r\cdot k}\omega_k + b\right) \tag{1}$$

where y_i represents the result of the *ith* component, p_i is the input value of the *ith* peptide, ω indicates the parameter of the filter, and K stands for the size of the kernel. The dilation rate, denoted as r, determines the spacing between kernel elements, and when $r = 1$ in the dilation convolution, it is equivalent to the standard convolution. When the dilation rate is 2, a 0 is inserted into the adjacent convolution weights. The first stage of this component entails feeding the matrix that represents the embedded structure of the peptide sequence into parallel convolutional units. These units are equipped with unique dilation rates (specifically, 2, 4, and 8) to capture diverse aspects of the peptide chain across multiple levels. Following that, to address the issue of overfitting and derive the convolutional feature matrix, a maximum pooling operation is employed.

2.4.1.2. Bi-LSTM. The Recurrent Neural Network (RNN) captures the complete historical context of the input data [52]. However, RNNs exhibit certain limitations, such as the potential for vanishing or exploding gradients during backpropagation. Within the DeepACP [53] framework, three discrete deep learning models were introduced, and among them, the RNN model exhibited the utmost accuracy in predicting ACPs. Long Short-Term Memory (LSTM) [54], a modified form of RNN, addresses these challenges by incorporating input, output, and forget gates. The fundamental concept underlying LSTM is that multiple gates regulate the flow of information over time, enabling the extraction of relevant information at each time step. ACP-DL [55] introduced a deep learning model based on LSTM, which has demonstrated commendable performance in automatic anticancer peptide feature extraction. Bi-LSTM [56] is a variant of RNN that consists of two LSTM layers combined together. It allows encoding of the information from back to front and calculates a forgetting gate cell f_t as:

$$f_t = \sigma\left(W_f\cdot[h_{t-1}, x_t] + b_f\right), \tag{2}$$

$$v_t = \tanh(W_c\cdot[h_{t-1}, x_t] + b_v), \tag{3}$$

where σ is a nonlinear activation function in the range of [0, 1] to introduce nonlinearity into the model. The training matrix is represented by the weight W and bias b. h indicates a hidden layer cell. f_t represents the forget gate unit, while v_t corresponds to the cell state unit.

The input state is calculated as follows:

$$i_t = \sigma(W_i\cdot[h_{t-1}, x_t] + b_i), \tag{4}$$

additionally, cell status is computed as follows:

$$C_t = f_t\cdot C_{t-1} + i_t\cdot v_t, \tag{5}$$

finally, the synchronization gate o_t is responsible for coordinating the information from the previous cell and producing the output for the current time as:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o), \tag{6}$$

Hidden state h_t is obtained as:

$$h_t = o_t\cdot\tanh(C_t), \tag{7}$$

the cell state C_t is expressed by the cell state cell v_t, which corresponds to the update gate cell, also known as the input cell.

LSTM can capture long-range dependencies in bioactive peptides very well. However, in the prediction framework, the LSTM is unable to capture the information in a reverse sequence, meaning that it does not incorporate future information. Compared to the standard LSTM, the Bi-LSTM model has the ability to extract significance from both past and present information so the network can better understand the contextual

information of bioactive peptides. The comprehensive architecture of the Bi-LSTM hidden module, incorporating concatenated vectors for both the forward and reverse processes, is illustrated below:

$$h_t = \overrightarrow{h^t} \oplus \overleftarrow{h^t}, \tag{8}$$

where the symbol $\oplus$ represents element-wise summation, which is utilized to combine the outputs from the forward and reverse processes.

2.4.2. Multi-layer convolution branch

Convolutional neural networks (CNN), among the prominent techniques in the field of deep neural networks, are a type of feedforward neural network that incorporates convolutional processing. CNN's end-to-end architecture makes it a powerful tool for bioactive peptides feature extraction, incorporating a feature extraction mechanism that comprises a convolutional layer and a pooling layer. This localized connectivity scheme empowers the network to capture relevant bioactive peptides features efficiently while minimizing computational overhead and model complexity. We employ a multi-layer convolutional model as the multi-layer convolution branch to extract the structural key of multi-functional bioactive peptides as:

$$X = b_j^k + \sum_{i=1}^{N_{k-1}} \mathrm{conv1D}\left(x_i^{k-1}, w_{ij}^{k-1}\right), \tag{9}$$

where the number of filters is 16, 32, and 32, respectively.

Once the convolutional calculation is terminated, the filtering outcomes for every level undergo non-linear treatment with the activation function. Specifically, the Rectified Linear Unit (ReLU) activation function is employed to process the outputs of convolutions as:

$$Y = \mathrm{Relu}(\mathrm{Relu}(\mathrm{Relu}(X))), \tag{10}$$

2.4.3. Classification module

In our study, we utilize a fully connected layer as the prediction layer, seamlessly integrating the outcome of two branches. This unified feature is then fed as the primary input to the output layer, facilitating the prediction process. In the context of multi-label problems, we employ the binary cross-entropy loss function to treat the probabilities assigned to each node as independent variables. Specifically, the binary cross-entropy loss function is defined as follows:

$$\begin{aligned}\mathrm{BCELoss} &= -\frac{1}{N}\sum_{i=1}^{N}(y_i \log(p(y_i))) \\ &\quad + (1-y_i)\log(1-p(y_i)))),\end{aligned} \tag{11}$$

where y_i represents the label as 1 represents the presence of this function in the peptide, while 0 indicates its absence. The terms $p(y_i)$ and $1 - p(y_i)$ correspond to the predicted probabilities for 0 and 1 of the *ith* sample, respectively, while N represents the total number of samples. Sigmoid activation function was employed to obtain scores for each node within the range of 0–1. Subsequently, a cutoff point of 0.5 is utilized to determine the predicted classes for each category. The prediction vector generated consists of five dimensions, with each dimension corresponding to a specific category, namely ACP, ADP, AHP, AIP, and AMP.

2.4.4. Evaluation metrics

We partition the tested peptides into two sets: the training dataset, which encompassed 80% of the samples, and the validation dataset, which accounted for the remaining 20%. The neural network underwent training utilizing the training dataset and subsequent evaluation on the validation dataset. To facilitate meaningful comparisons with state-of-the-art methodologies, we employ performance metrics similar to those utilized in the MLBP framework. These metrics are thoughtfully designed to gauge the efficacy and predictive performance as:

$$\mathrm{Precision} = \frac{1}{N}\sum_{i=1}^{N}\frac{\left\|L_i \cap L_i^*\right\|}{\left\|L_i^*\right\|}, \tag{12}$$

where the set L_i stands for the actual labels assigned to sample I, while L_i^* corresponds to the predicted labels for the same sample. The variable N represents the total count of test samples in the dataset. $\cap$ indicates the concatenation of sets.

$$\mathrm{Coverage} = \frac{1}{N}\sum_{i=1}^{N}\frac{\left\|L_i \cap L_i^*\right\|}{\|L_i\|}, \tag{13}$$

$$\mathrm{Accuracy} = \frac{1}{N}\sum_{i=1}^{N}\frac{\left\|L_i \cap L_i^*\right\|}{\left\|L_i \cup L_i^*\right\|}, \tag{14}$$

where $\cup$ indicates the intersection of sets.

$$\mathrm{Absolute\ False} = \frac{1}{N}\sum_{i=1}^{N}\frac{\left\|L_i \cup L_i^*\right\| - \left\|L_i \cap L_i^*\right\|}{M}, \tag{15}$$

where M represents the functional category of the low-redundancy dataset.

$$\mathrm{Absolute\ True} = \frac{1}{N}\sum_{i=1}^{N}\Delta\left(L_i, L_i^*\right), \tag{16}$$

where Δ is described as:

$$\Delta\left(L_i, L_i^*\right) = \begin{cases} 1, & \text{if } L_i^* \text{ is identical to } L_i \\ 0 & \text{other} \end{cases} \tag{17}$$

In terms of Precision, Coverage, Accuracy, and Absolute True, larger values mean better prediction performance. Conversely, fewer Absolute False indicate better prediction performance.

3. Experiments and analysis

We demonstrate the performance of our proposed MMDB, which integrates two distinct feature extraction branches: multi-scale dilated convolution with Bi-LSTM branch and multi-layer convolution branch. In order to demonstrate the effectiveness of different branches, we conduct ablation experiments on two branches respectively. Compared to the state-of-the-art methods, Precision, Coverage, and Accuracy indicators are improved by 2.6%, 6.4%, and 0.85% respectively by peptide multi-scale dilated convolution with Bi-LSTM branch. Meanwhile, Precision, Coverage, Accuracy, and Absolute True obtain with more gains of 2.6%, 2.5%, 2.7%, and 2.9%, respectively by multi-layer convolution branch that effectively incorporates structural information.

3.1. Comparison with competing methods

To assess the efficacy of the MMDB approach, we perform a comparative analysis with five state-of-the-art methods, which include one deep learning-based algorithm and four machine learning-based algorithms.

- MLBP [31] (2022) serves as a benchmark by utilizing CNN + Bi-GRU for feature extraction and sequence fusion. However, it solely relies on the sequence input of active peptides and ignores the structural key.
- RBRL [57] (2020) is an algorithm based on low-rank learning that tackles the multi-label classification problem by transforming it into multiple single-label classification problems and utilizes low-rank decomposition to reduce data dimensionality and computational complexity.

- MLDF [58] (2019) handles large-scale multi-label classification problems by leveraging multiple random forests to predict the probability of each label, followed by the use of the softmax function to normalize these probabilities.
- CLR [59] (2008) addresses the multilabel classification challenge by implementing a second-order algorithm to convert it into multiple binary classification problems.
- RAKEL [60] (2007) divides the labels into several subsets before transforming them into binary classification problems and acts as a high-order algorithm.

3.2. Experimental results

As demonstrated in Table 1, MMDB obviously outperforms state-of-the-art methods on all metrics. In comparison to CLR, which exhibits the highest performance in the machine learning-based multi-classification algorithms on five crucial metrics, MMDB achieves 9%, 13%, 10%, 7%, and 23% improvements. As shown in Fig. 4, our approach yields the lowest value for the Absolute False metric as 0.102%. A comprehensive evaluation between MMDB and MLBP exhibits the highest performance in the deep learning-based multi-classification algorithms, providing a thorough evaluation. Specifically, our method demonstrates superior performance with the largest performance improvement on Coverage at 9.1%, and the average improvement on Precision, Accuracy, and Absolute True is nearly 3.8%. A comprehensive evaluation of the performance of MMDB and MLBP exhibits the highest performance in the deep learning-based multi-classification algorithms. As illustrated in Fig. 5, our method demonstrates superior performance with the improvement in Coverage as 9.1%, on Precision and Accuracy as 5.3% and 3.5%.

3.3. Analysis

3.3.1. Evaluation of multi-scale dilated convolution with Bi-LSTM branch

To validate the performance of multi-scale dilated convolution with Bi-LSTM branch, we conduct experiments on the amino acid sequence vector by removing the multi-layer convolution branch. In accordance with standard machine learning practices, the dataset is divided into a training set (80%) and a test set (20%) through random partitioning. As shown in Table 2, this branch exhibits noTable improvements over MLBP on most metrics except the Absolute True metric. Specifically, in comparison with MLBP, we observe an enhancement of 2.6% in Precision, 6.4% in Coverage, and 0.85% in Accuracy. The results illustrate that the multi-scale dilated convolution and Bi-LSTM model exhibits strong sequence vector extraction capabilities.

3.3.1.1. The effects of multi-scale dilated convolution. For further analysis of the different components of multi-scale dilated convolution with Bi-LSTM branch, we perform ablation experiments to assess the performance of multi-scale dilated convolution for extracting peptide sequence features. The result in Table 3 indicates that multi-scale dilated convolution significantly outperforms normal multi-scale CNN in extracting peptide features since the dilated convolution technique can effectively capture diverse-scale contextual knowledge by significantly enlarging the effective receptive region. As the length range of multi-functional bioactive peptides varies from 5 to 517 residues, we make use of different dilated convolution rates to extract long-range dependence from peptide sequences. To investigate the effect of different dilated rates, we then set the dilated rates of the three convolution blocks in the multi-scale convolution block to 2, 4, and 8, respectively. The results of different dilated rates on the benchmark data set are shown in Fig. 3 as (a), (b), (c), (d), and (e) illustrate the performance of different dilation rates on five evaluation metrics such as s Precision, Coverage, Accuracy, Absolute True, and Absolute False. This approach achieves significantly better results, especially with r(2,4,8), than setting the same dilated rates for all convolution blocks. It is noted that all dilated rates are set to the same value and yield similar results but fail to extract features from peptides that are either too long or too short.

Multi-scale dilation convolution with different rates for various length ranges effectively extracts features of different length spans since the selective skipping of input neurons during convolution enlarges the receptive field and enables the network to extract diverse features from a wider span of peptide sequences. The implementation of multi-scale dilation convolution provides a promising approach by capturing the complex interactions between amino acid residues and modeling the underlying mechanisms of peptide function. These findings carry significant implications for the development of more accurate and reliable models for predicting the functional properties of multi-functional bioactive peptides, which is a critical task in bioinformatics and biomedicine.

Table 1
Performance of feature fusion models based on multimodal data compared with optimal methods.

Method	Precision ↑	Coverage ↑	Accuracy ↑	Absolute True ↑	Absolute False ↓
RBRL	0.650	0.651	0.649	0.646	0.140
MLDF	0.649	0.649	0.648	0.646	0.119
RAKEL	0.649	0.648	0.648	0.647	0.141
CLR	0.667	0.677	0.666	0.655	0.133
MLBP	0.697	0.701	0.708	0.699	0.106
MMDB	**0.734**	**0.765**	**0.733**	**0.701**	**0.102**

Table 2
Performance of the multi-scale dilated convolution and Bi-LSTM model compared with MLBP.

Method	Precision ↑	Coverage ↑	Accuracy ↑	Absolute True ↑	Absolute False ↓
MLBP	0.697	0.701	0.708	**0.699**	**0.106**
Multi-scale dilated convolution with Bi-LSTM branch	**0.715**	**0.746**	**0.714**	0.681	**0.106**

3.3.1.2. The effects of Bi-LSTM. The Bi-LSTM module in our proposed model serves to integrate and synthesize the multi-scale local features extracted from the dilated convolution process effectively. In order to assess the contribution of the Bi-LSTM module to the overall effectiveness of the approach, we design and evaluate five different deep neural network (DNN) architectures on a test set. The performance of each DNN is compared, and Bi-LSTM achieves the peak of prediction precision, as indicated by the results. A detailed summary of the performance metrics for each of the five DNN models is provided in Table 4. These findings support the critical role played by the Bi-LSTM module in enhancing the overall feature extraction and prediction capabilities.

3.3.2. Evaluation of multi-layer convolution branch

To investigate the significance of the multi-layer convolution branch of the MMDB model, we design a set of experiments by five different metrics under two different input configurations: (1) when only the multi-scale dilated convolution with Bi-LSTM branch is retained, and (2) when both the multi-scale dilated convolution with Bi-LSTM branch and the multi-layer convolution branch are both retained. The experimental results (Table 5, Fig. 6) reveal that the performance of the MMDB model is consistently better under diverse input configurations. Specifically, the MMDB model shows significant improvements of 2.6%, 2.5%, and 2.7% on the Precision, Coverage, and Accuracy metrics, respectively. The comparison on the Absolute True metric also shows a promising 2.9% increase with a better alignment between the predicted and actual functional classifications of the peptides. These results suggest that the

Table 3
Comparison of the performance of different dilated rates (r) on the benchmark data set.

Method	Precision ↑	Coverage ↑	Accuracy ↑	Absolute True ↑	Absolute False ↓
r(1,1,1)	0.717	0.744	0.717	0.689	0.108
r(2,2,2)	0.723	0.745	0.722	0.698	0.105
r(4,4,4)	0.720	0.745	0.719	0.691	0.107
r(8,8,8)	0.724	0.748	0.723	0.697	0.105
r(2,4,8)	**0.734**	**0.765**	**0.733**	**0.701**	**0.102**

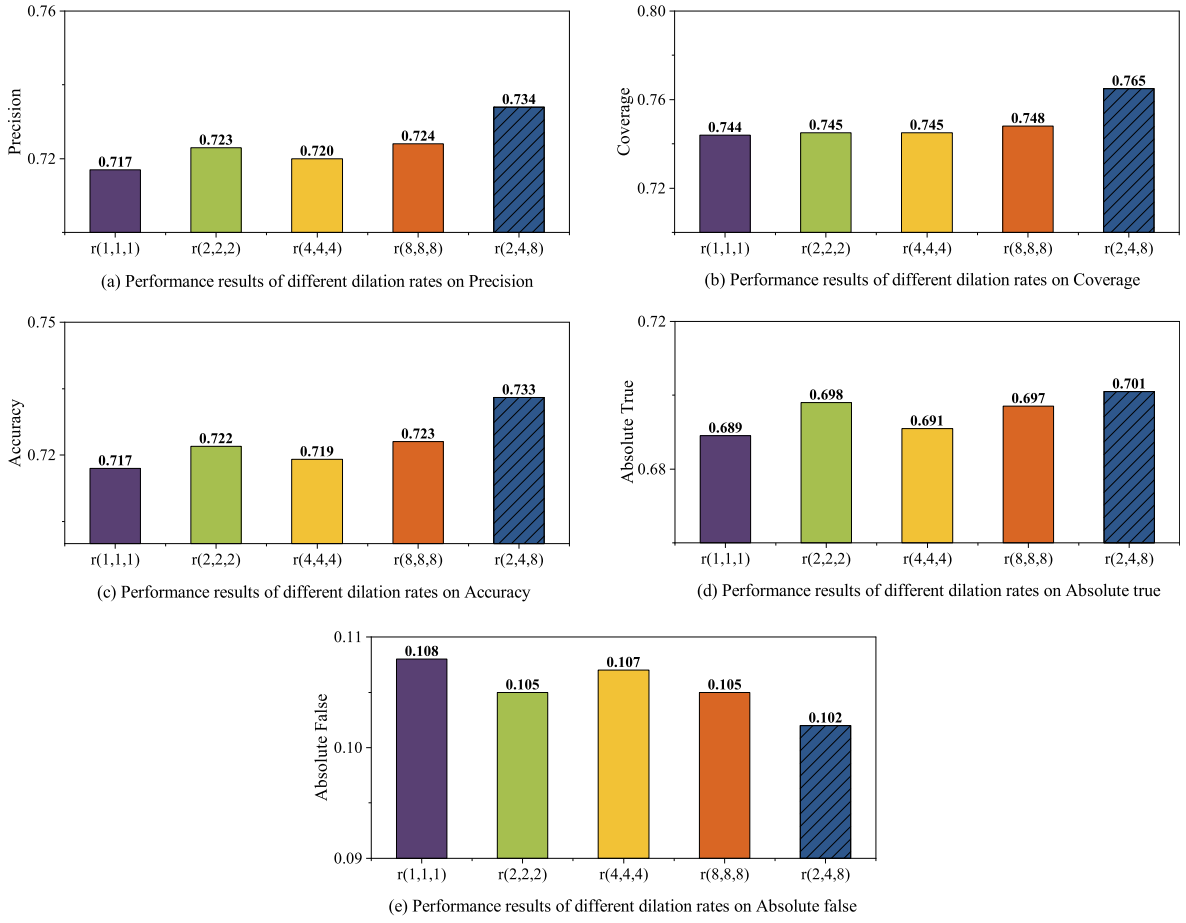


Fig. 3. Performance of different dilated rates on the benchmark data set. (a), (b), (c), (d), and (e) illustrate the performance results of different dilation rates on five evaluation metrics: Precision, Coverage, Accuracy, Absolute True, and Absolute False. The term r(α,β,γ) represents different dilation rates on multiple scales, where α, β, and γ denote the dilation rate. Specifically, r(1,1,1) indicates that each scale employs a dilation rate of 1, representing a regular convolution operation.

incorporation of the structural key into the model can enhance its ability to predict the intracellular biological function of peptides.

4. Conclusions

This study presents a novel lightweight approach, MMDB, for addressing the multi-functional bioactive peptide multi-label prediction problem. The MMDB algorithm designs a multi-branch structure to extract features from multi-modal peptide sequences and structural representations, resulting in significantly improved prediction performance. Specifically, our approach involves training the peptide embeddings of each modality separately by different branches and then integrating the information together. In particular, dilated convolution and Bi-LSTM encoder can capture the multi-scale sequence feature effectively. The proposed model outperforms the state-of-the-art methods in terms of five crucial metrics, demonstrating its potential as an effective tool for identifying, annotating, and characterizing novel multi-functional bioactive peptides.

Owing to constraints within the dataset, comprising a limited quantity and variety of peptides, along with challenges related to data imbalance, the enhancement of prediction accuracy has been impeded. Henceforth, we plan to further refine our model by exploring more effective modalities for addressing constraints related to the limited quantity and variety of peptides. Additionally, we will continue to optimize the model architecture by taking advantage of complementary between diverse modalities of peptide, with the aim of further improving the prediction performance of MMDB and alleviating with data imbalance.

CRediT authorship contribution statement

Yan Kang: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Huadong Zhang:** Data curation, Software, Writing – original draft, Writing – review & editing. **Xinchao Wang:** Visualization. **Yun Yang:** Supervision. **Qi Jia:** Investigation.

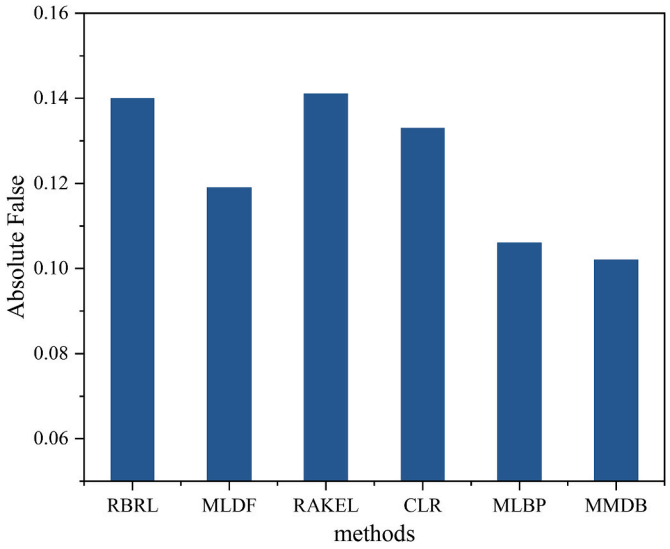


Fig. 4. Performance of MMDB and previous methods. MMDB is significantly better than previous methods in Absolute False.

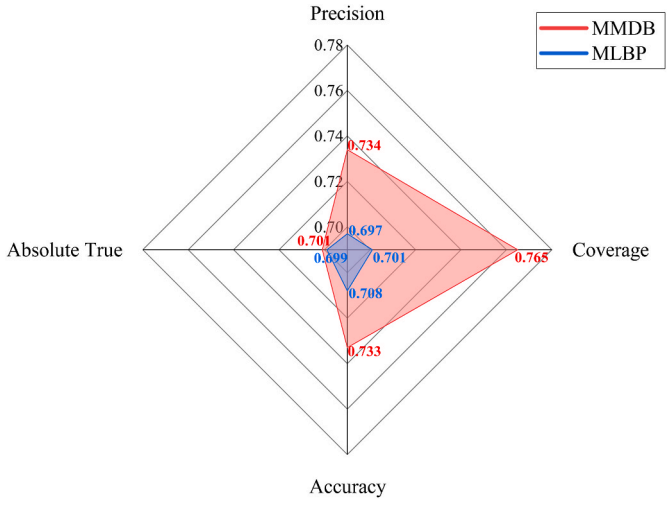


Fig. 5. Performance of MMDB and MLBP. MMDB is significantly better than MLBP in Precision, Coverage, Accuracy, and Absolute True.

Table 4
Performance of Bi-LSTM versus other DNN models on benchmark datasets.

Method	Precision ↑	Coverage ↑	Accuracy ↑	Absolute True ↑	Absolute False ↓
Without Bi-LSTM	0.677	0.696	0.677	0.656	0.111
GRU	0.715	0.741	0.714	0.686	0.104
Bi-GRU	0.726	0.750	0.725	0.699	0.102
LSTM	0.722	0.754	0.721	0.687	0.105
Bi-LSTM	**0.734**	**0.765**	**0.733**	**0.701**	**0.102**

Table 5
Validation results of multi-layer convolution branch.

Method	Precision ↑	Coverage ↑	Accuracy ↑	Absolute True ↑	Absolute False ↓
Sequence branch	0.715	0.746	0.714	0.681	0.106
Two branches	**0.734**	**0.765**	**0.733**	**0.701**	**0.102**

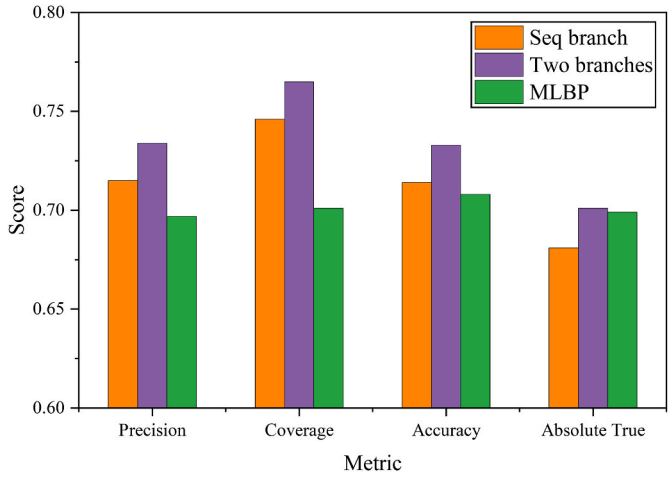


Fig. 6. Performance of two branches of MMDB and MLBP.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China (Grant No. 62366060, 61762092); Open Foundation of Yunnan Key Laboratory of Software Engineering (Grant No.2023SE203); the Major Science and Technology Projects in Yunnan Province under Grant No. 202002AD080047 and 202202AE090019; and the Postgraduate Research and Innovation Foundation of Yunnan University (Grant No. 2021Y407).

References

[1] Eric Banan-Mwine Daliri, Byong H. Lee, Deog H. Oh, Current trends and perspectives of bioactive peptides, Crit. Rev. Food Sci. Nutr. 58 (13) (2018) 2273–2284.
[2] Peng Qi, Yingwen Wang, Dun Zhang, Yan Sun, Laibao Zheng, Multichannel bacterial discrimination based on recognition and disintegration disparity of short antimicrobial peptides, Anal. Biochem. 600 (2020) 113764.
[3] Bilal Cakir, Tugba Tunali-Akbay, Potential anticarcinogenic effect of goat milk-derived bioactive peptides on hct-116 human colorectal carcinoma cell line, Anal. Biochem. 622 (2021) 114166.
[4] Min Kyoung Shin, Byungjo Lee, Seung Tae Kim, Yoo Jung Sun, Jung-Suk Sung, Designing a novel functional peptide with dual antimicrobial and anti-inflammatory activities via in silico methods, Front. Immunol. 13 (2022) 821070.
[5] Kazuhiko Tatemoto, Kiyoshige Takayama, Min-Xu Zou, Iku Kumaki, Wei Zhang, Kimitsuka Kumano, Mineko Fujimiya, The novel peptide apelin lowers blood pressure via a nitric oxide-dependent mechanism, Regul. Pept. 99 (2–3) (2001) 87–92.
[6] Jaspreet Kaur, Vikas Kumar, Kartik Sharma, Sawinder Kaur, Yogesh Gat, Ankit Goyal, Beenu Tanwar, Opioid peptides: an overview of functional significance, Int. J. Pept. Res. Therapeut. 26 (2020) 33–41.
[7] Adrián Sánchez, Alfredo Vázquez, Bioactive peptides: a review, Food Qua. Safet. 1 (1) (2017) 29–46.
[8] Luca Domenico D'Andrea, Alessandra Romanelli, Temporins: multifunctional peptides from frog skin, Int. J. Mol. Sci. 24 (6) (2023) 5426.
[9] Ning Wang, Jun Zhang, Bin Liu, Idrbp-ppct: identifying nucleic acid-binding proteins based on position-specific score matrix and position-specific frequency matrix cross transformation, IEEE ACM Trans. Comput. Biol. Bioinf 19 (4) (2021) 2284–2293.
[10] Xiangzheng Fu, Lijun Cai, Xiangxiang Zeng, Quan Zou, Stackcppred: a stacking and pairwise energy content-based prediction of cell-penetrating peptides and their uptake efficiency, Bioinformatics 36 (10) (2020) 3028–3034.

[11] Ke Yan, Hongwu Lv, Yichen Guo, Yongyong Chen, Hao Wu, Bin Liu, Tppred-atmv: therapeutic peptide prediction by adaptive multi-view tensor learning model, Bioinformatics 38 (10) (2022) 2712–2718.
[12] Yu P. Zhang, Quan Zou, Pptpp: a novel therapeutic peptide prediction method using physicochemical property encoding and adaptive feature representation learning, Bioinformatics 36 (13) (2020) 3982–3987.
[13] Fereshteh Fallah Atanaki, Saman Behrouzi, Shohreh Ariaeenejad, Boroomand Amin, Kaveh Kavousi, Bipep: sequence-based prediction of biofilm inhibitory peptides using a combination of nmr and physicochemical descriptors, ACS Omega 5 (13) (2020) 7290–7297.
[14] Vishakha Singh, Sameer Shrivastava, Sanjay Kumar Singh, Abhinav Kumar, Sonal Saxena, Stable-abppred: a stacked ensemble predictor based on bilstm and attention mechanism for accelerated discovery of antibacterial peptides, Briefings Bioinf. 23 (1) (2022) bbab439.
[15] Bing Rao, Chen Zhou, Guoying Zhang, Ran Su, Leyi Wei, Acpred-fuse: fusing multi-view information improves the prediction of anticancer peptides, Briefings Bioinf. 21 (5) (2020) 1846–1855.
[16] Cui-Fang Gao, Zi-Xue Qiu, Xiao-Jun Wu, Feng-Wei Tian, Hao Zhang, Wei Chen, A novel fuzzy Fisher classifier for signal peptide prediction, Protein Pept. Lett. 18 (8) (2011) 831–838.
[17] Balachandran Manavalan, Tae Hwan Shin, Myeong Ok Kim, Gwang Lee, Pip-el: a new ensemble learning method for improved proinflammatory peptide predictions, Front. Immunol. 9 (2018) 1783.
[18] Atul Tyagi, Pallavi Kapoor, Rahul Kumar, Kumardeep Chaudhary, Ankur Gautam, G.P.S. Raghava, In silico models for designing and discovering novel anticancer peptides, Sci. Rep. 3 (1) (2013) 1–8.
[19] Abu Zahid Bin Aziz, Md Al Mehedi Hasan, Shamim Ahmad, Md Al Mamun, Jungpil Shin, Md Rahat Hossain, iacp-multicnn: multi-channel cnn based anticancer peptides identification, Anal. Biochem. 650 (2022) 114707.
[20] Sen Yang, Zexi Yang, Xinye Ni, Ampfinder: a computational model to identify antimicrobial peptides and their functions based on sequence-derived information, Anal. Biochem. 673 (2023) 115196.
[21] Ritesh Sharma, Sameer Shrivastava, Sanjay Kumar Singh, Abhinav Kumar, Sonal Saxena, Raj Kumar Singh, Deep-abppred: identifying antibacterial peptides in protein sequences using bidirectional lstm with word2vec, Briefings Bioinf. 22 (5) (2021) bbab065.
[22] Shuihua Wang, M. Emre Celebi, Yu-Dong Zhang, Yu Xiang, Siyuan Lu, Xujing Yao, Qinghua Zhou, Miguel Martinez-Garcia, Yingli Tian, Juan M. Gorriz, et al., Advances in data preprocessing for biomedical data fusion: an overview of the methods, challenges, and prospects, Inf. Fusion 76 (2021) 376–421.
[23] Yu-Dong Zhang, Zhengchao Dong, Shui-Hua Wang, Yu Xiang, Xujing Yao, Qinghua Zhou, Hua Hu, Min Li, Carmen Jiménez-Mesa, Javier Ramirez, et al., Advances in multimodal data fusion in neuroimaging: overview, challenges, and novel orientation, Inf. Fusion 64 (2020) 149–187.
[24] Sören Richard Stahlschmidt, Benjamin Ulfenborg, Jane Synnergren, Multimodal deep learning for biomedical data fusion: a review, Briefings Bioinf. 23 (2) (2022) bbab569.
[25] Luca Bonomi, Liyue Fan, Xiaoqian Jiang, Noise-tolerant similarity search in temporal medical data, J. Biomed. Inf. 113 (2021) 103667.
[26] Ivan Izonin, Roman Tkachenko, Khrystyna Zub, Pavlo Tkachenko, et al., A grnn-based approach towards prediction from small datasets in medical application, Procedia Comput. Sci. 184 (2021) 242–249.
[27] Shahla Faisal, Gerhard Tutz, Imputation methods for high-dimensional mixed-type datasets by nearest neighbors, Comput. Biol. Med. 135 (2021) 104577.
[28] Chunwei Tian, Yong Xu, Lunke Fei, Junqian Wang, Jie Wen, Nan Luo, Enhanced cnn for image denoising, CAAI Transact. Intelligen. Tech. 4 (1) (2019) 17–23.
[29] Vegard Antun, Francesco Renna, Clarice Poon, Ben Adcock, Anders C. Hansen, On instabilities of deep learning in image reconstruction and the potential costs of ai, Proc. Natl. Acad. Sci. USA 117 (48) (2020) 30088–30095.
[30] Sjoerd Van Steenkiste, Karol Kurach, Jürgen Schmidhuber, Sylvain Gelly, Investigating object compositionality in generative adversarial networks, Neural Network. 130 (2020) 309–325.
[31] Wending Tang, Ruyu Dai, Wenhui Yan, Wei Zhang, Bin Yannan, Enhua Xia, Junfeng Xia, Identifying multi-functional bioactive peptide functions using multi-label deep learning, Briefings Bioinf. 23 (1) (2022) bbab414.
[32] Yu Fisher, Vladlen Koltun, Multi-scale Context Aggregation by Dilated Convolutions, 2015 *arXiv preprint arXiv:1511.07122*.
[33] Ali Adem Bahar, Dacheng Ren, Antimicrobial peptides, Pharmaceuticals 6 (12) (2013) 1543–1575.
[34] Lath Amit, Anita Rani Santal, Nameet Kaur, Poonam Kumari, Nater Pal Singh, Anti-cancer peptides: their current trends in the development of peptide-based therapy and anti-tumor drugs, Biotechnol. Genet. Eng. Rev. 39 (1) (2023) 45–84.
[35] Caleb Acquah, Christian Ko Dzuvor, Susan Tosh, Dominic Agyei, Anti-diabetic effects of bioactive peptides: recent advances and clinical implications, Crit. Rev. Food Sci. Nutr. 62 (8) (2022) 2158–2171.
[36] Rotimi E. Aluko, Antihypertensive peptides from food proteins, Annu. Rev. Food Sci. Technol. 6 (2015) 235–262.
[37] Ying Luo, Yuzhu Song, Mechanism of antimicrobial peptides: antimicrobial, anti-inflammatory and antibiofilm activities, Int. J. Mol. Sci. 22 (21) (2021) 11401.
[38] Mst Khatun, Md Hasan, Hiroyuki Kurata, et al., Preaip: computational prediction of anti-inflammatory peptides by integrating multiple complementary features, Front. Genet. 10 (2019) 129.
[39] Balachandran Manavalan, Shaherin Basith, Tae Hwan Shin, Leyi Wei, Gwang Lee, mahtpred: a sequence-based meta-predictor for improving the prediction of anti-hypertensive peptides using effective feature representation, Bioinformatics 35 (16) (2019) 2757–2765.
[40] Susanta Roy, Robindra Teron, Biodadpep: a bioinformatics database for anti diabetic peptides, Bioinformation 15 (11) (2019) 780.
[41] Piyush Agrawal, Dhruv Bhagat, Manish Mahalwal, Neelam Sharma, Gajendra Ps Raghava, Anticp 2.0: an updated model for predicting anticancer peptides, Briefings Bioinf. 22 (3) (2021) bbaa153.
[42] Chia-Ru Chung, Ting-Rung Kuo, Li-Ching Wu, Tzong-Yi Lee, Horng Jorng-Tzong, Characterization and identification of antimicrobial peptides with different functional activities, Briefings Bioinf. 21 (3) (2020) 1098–1114.
[43] Shuhui Zhang, Lu Luo, Xueyan Sun, Aimin Ma, Bioactive peptides: a promising alternative to chemical preservatives for food preservation, J. Agric. Food Chem. 69 (42) (2021) 12369–12384.
[44] Md Ashraf Hussain, Tofael Ahmed Sumon, Sabuj Kanti Mazumder, Mir Mohammad Ali, Won Je Jang, Muyassar Hamid Abualreesh, S.M. Sharifuzzaman, Christopher Lyon Brown, Hyun-Tai Lee, Eun-Woo Lee, et al., Essential oils and chitosan as alternatives to chemical preservatives for fish and fisheries products: a review, Food Control 129 (2021) 108244.
[45] Kaustav Majumder, Jianping Wu, Molecular targets of antihypertensive peptides: understanding the mechanisms of action based on the pathophysiology of hypertension, Int. J. Mol. Sci. 16 (1) (2014) 256–283.
[46] Sudheer Gupta, Ashok K. Sharma, Vibhuti Shastri, Midhun K. Madhu, Vineet K. Sharma, Prediction of anti-inflammatory proteins/peptides: an insilico approach, J. Transl. Med. 15 (1) (2017) 1–11.
[47] Mingfeng Xie, Dijia Liu, Yufeng Yang, Anti-cancer peptides: classification, mechanism of action, reconstruction and modification, Open Biol. 10 (7) (2020) 200004.
[48] Jing Zhao, Lingchuang Bai, Xiang-kui Ren, Jintang Guo, Shihai Xia, Wencheng Zhang, Yakai Feng, Co-immobilization of ach11 antithrombotic peptide and cag cell-adhesive peptide onto vascular grafts for improved hemocompatibility and endothelialization, Acta Biomater. 97 (2019) 344–359.
[49] Yan Han, Baoping Tang, Lei Deng, An enhanced convolutional neural network with enlarged receptive fields for fault diagnosis of planetary gearboxes, Comput. Ind. 107 (2019) 50–58.
[50] Liang-Chieh Chen, Papandreou George, Iasonas Kokkinos, Kevin Murphy, Alan L. Yuille, Deeplab: semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs, IEEE Trans. Pattern Anal. Mach. Intell. 40 (4) (2017) 834–848.
[51] Yan Chen, Qian Guo, Xinyan Liang, Jiang Wang, Yuhua Qian, Environmental sound classification with dilated convolutions, Appl. Acoust. 148 (2019) 123–132.
[52] Yong Yu, Xiaosheng Si, Changhua Hu, Jianxun Zhang, A review of recurrent neural networks: lstm cells and network architectures, Neural Comput. 31 (7) (2019) 1235–1270.
[53] Lezheng Yu, Runyu Jing, Fengjuan Liu, Jiesi Luo, Yizhou Li, Deepacp: a novel computational approach for accurate identification of anticancer peptides by deep learning algorithm, Mol. Ther. Nucleic Acids 22 (2020) 862–870.
[54] Sepp Hochreiter, Jürgen Schmidhuber, Long short-term memory, Neural Comput. 9 (8) (1997) 1735–1780.
[55] Hai-Cheng Yi, Zhu-Hong You, Xi Zhou, Li Cheng, Li Xiao, Tong-Hai Jiang, Zhan-Heng Chen, Acp-dl: a deep learning long short-term memory model to predict anticancer peptides using high-efficiency feature representation, Mol. Ther. Nucleic Acids 17 (1–9) (2019).
[56] Mike Schuster, K Paliwal Kuldip, Bidirectional recurrent neural networks, IEEE Trans. Signal Process. 45 (11) (1997) 2673–2681.
[57] Guoqiang Wu, Ruobing Zheng, Yingjie Tian, Dalian Liu, Joint ranking svm and binary relevance with robust low-rank learning for multi-label classification, Neural Network. 122 (2020) 24–39.
[58] Liang Yang, Xi-Zhu Wu, Yuan Jiang, Zhi-Hua Zhou, Multi-label Learning with Deep Forest, 2019 *arXiv preprint arXiv:1911.06557*.
[59] Johannes Fürnkranz, Eyke Hüllermeier, Eneldo Loza Mencía, Klaus Brinker, Multilabel classification via calibrated label ranking, Mach. Learn. 73 (2008) 133–153.
[60] Grigorios Tsoumakas, Ioannis Vlahavas, Random k-labelsets: an ensemble method for multilabel classification, in: Machine Learning: ECML 2007: 18th European Conference on Machine Learning, Warsaw, Poland, September 17-21, 2007. Proceedings 18, Pages 406–417. Springer, 2007.